ROOMAN
INNOVATE INTEGRATE EMPOWER

INTERNSHIP AT

ROOMAN TECHNOLOGIES

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COURSE STRUCTURE

ROOMAN
INNOVATE INTEGRATE EMPOWER

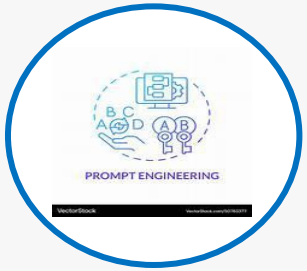
1. IT IGNITE
2. AI IN CYBER SECURITY
3. PROJECT

IT IGNITE

Week 1

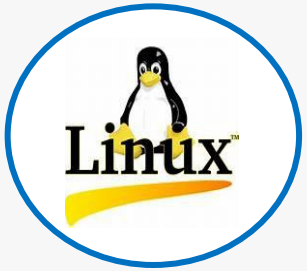
Python Programming

Week 2



Prompt Engineering

Week 3



Linux Basics

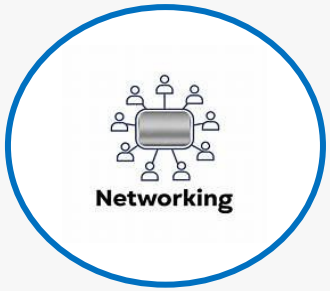
Week 4

AI Coding Assistants & Modern Dev Workflows

COURSE STRUCTURE

IT IGNITE

Week 5



Networking Essentials

Week 6

Cloud Fundamentals

Week 7

Cyber Security

Week 8

Database

Week 9

Software Engineering

Week 10

Version Control &Github

Week 11

Digital Literacy

Week 12

Workplace Readiness

WEEK-WISE TIMELINE

WEEK -7

- NoSQL Databases
- Modern Data Storage
- Connecting Applications to Databases
- IPv4

WEEK -8

- Software Engineering Concepts & System Types.
- Networking Basics (OSI, Devices, Subnetting).
- SDLC Models (Waterfall, Agile, DevOps).
- Networking Design & Ipv6.

WEEK -9

- Agile & Scrum Framework
- Firewalls, VPNs, IDS & IPS
- GRC Fundamentals
- Splunk Fundamentals

WEEK -10

- Version Control & Git.
- Staging, Committing, Undoing Changes.
- GitHub & Collaboration.
- Cisco CLI Modes & Commands.

WEEK -11

- VLAN Fundamentals
- VLANs, Trunks & VTP
- Routing Fundamentals & Network Protocols
- Routing Protocols (RIP, EIGRP, OSPF)
- Static Routing

WEEK -12

- Multilayer Switching (L2/L3)
- Router-on-a-Stick (ROAS)
- Data Types & Lifecycle
- Data Interpretation, Dashboards & KPIs
- Data Ethics & Privacy
- Data-Driven Decision Making & Case Studies.

TECHNICAL EXPOSURE OBTAINED DURING INTERNSHIP

DATABASE & DATA MANAGEMENT

NoSQL databases and modern data storage

Connecting applications to databases

Cloud databases and data handling

Data types and data lifecycle

Data interpretation, dashboards & KPIs

NETWORKING & INFRASTRUCTURE

IPv4 and IPv6 addressing | OSI model, subnetting & network design | VLAN fundamentals, trunks & VTP | Routing fundamentals and static routing | Routing protocols (RIP, EIGRP, OSPF) | Multilayer switching (L2/L3) | Router-on-a-Stick (ROAS)

SECURITY & GOVERNANCE

Firewalls, VPNs, IDS & IPS

GRC fundamentals (Governance, Risk, Compliance)

Data ethics and privacy regulations

Cybersecurity monitoring using Splunk

DEVELOPMENT & DEVOPS TOOLS

Agile and Scrum framework

SDLC models (Waterfall, Agile, DevOps)

Version control using Git & GitHub

Staging, committing & undoing changes

GitHub collaboration workflows

Cisco CLI modes and commands

SKILLS LEARNED DURING INTERNSHIP

TECHNICAL SKILLS

Database concepts (NoSQL, application connectivity) | Networking fundamentals (IPv4, IPv6, OSI model, subnetting) | VLANs, trunking, VTP and routing protocols | Static routing and multilayer switching (L2/L3) | Router-on-a-Stick (ROAS) configuration basics.

ANALYTICAL & PRACTICAL SKILLS

Understanding data lifecycle and data types | Data interpretation using dashboards & KPIs | Log analysis and monitoring using Splunk | Applying Agile and Scrum methodologies | Real-world problem solving using case studies.

SOFT SKILLS

Communication | Problem solving | Analytical thinking | Time management

Continuous learning through daily quizzes and assignments

TOOLS USED DURING INTERNSHIP

DATABASE & DATA TOOLS

MySQL – database management and querying
MongoDB – NoSQL database handling
Google Colab – data analysis and execution
Splunk – log monitoring and data analysis

VERSION CONTROL & COLLABORATION

Git – version control system
GitHub – code hosting and collaboration
Git Bash – command-line operations

NETWORKING & CONFIGURATION TOOLS

Cisco CLI – network device configuration
Packet Tracer – network simulation and design
Wireshark – packet capture and analysis

DEVELOPMENT & WORKFLOW TOOLS

Agile tools (Scrum boards) – project management
VS Code – code editing and development
Online compilers – MySQL and code execution

CONCLUSION

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- Developed a strong understanding of **database concepts**, including NoSQL and application-database connectivity.
- Gained knowledge of **networking fundamentals** such as IPv4, IPv6, OSI model, subnetting, and network design.
- Learned **advanced networking concepts** including VLANs, trunking, VTP, routing protocols, and static routing.
- Understood **software development practices** like SDLC models, Agile & Scrum methodologies, and DevOps basics.
- Acquired hands-on experience with **version control tools** like Git and GitHub for collaboration and code management.
- Explored **network security concepts** including firewalls, VPNs, IDS/IPS, and governance frameworks (GRC).
- Gained insights into **data concepts**, including data types, lifecycle, dashboards, KPIs, and data-driven decision making.
- Developed awareness of **data ethics, privacy regulations, and real-world case studies.**